JOSHUA DIMASAKA

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SUMMARY

Joshua Dimasaka is a geospatial data scientist and disaster risk researcher with over eight years of experience in earthquake and climate risk modelling, Earth observation, and deep probabilistic modelling. His multidisciplinary training spans civil and structural engineering, public policy, and environmental data science. From being the first Filipino Stanford Knight-Hennessy Scholar, he is now a UKRI-funded PhD candidate at the University of Cambridge, developing data-driven frameworks to quantify and audit disaster risk across Least Developed Countries, building on prior work at Stanford University, the German Aerospace Center, and the Earthquakes and Megacities Initiative. He has authored over six peer-reviewed publications, serves as an active scientific reviewer, and has translated his research into operational tools adopted by governments, industry, and international risk agencies. His strength in scientific communication has been recognized internationally at the European Geosciences Union (2026), the 8th International Disaster and Risk Conference (2025), and the American Geophysical Union (2023).

INTEREST

Geospatial Data Science, Earthquake Risk and Loss Modelling, Earth Observation Data,
Climate and Disaster Risk Assessment, Spatiotemporal Analysis, Deep Probabilistic Modelling

SKILLS

MATLAB, QGIS, Google Earth Engine, GitHub, Python, R, State, Tableau, LaTeX

EDUCATION

University of Cambridge — United Kingdom

- PhD Artificial Intelligence for the study of Environmental Risks — 2023-26
  Dissertation: Global Disaster Risk Quantification Audit using AI and Earth Observation Data
  Advisor: E. So (UCam), C. Geiß (DLR), R. Muir-Wood (Moody's RMS), F. Bendimerad (EMI)
- MRes Environmental Data Science — 2022-23
  Research: Exposure Assessment of Settlement-Road Systems from Climate-influenced Mass Movements
  Advisor: A. Marinoni (UiT), S. Selvakumaran (UCam)

Stanford University — United States

- MA Public Policy — 2020-22
  Research: Regional Earthquake Risk Assessment of the Greater Metro Manila Area, Philippines
  Advisor: J. Baker (Stanford)
- MS Civil and Environmental Engineering — 2019-22
  Specialization: Sustainable Design and Construction (Structural Engineering)
  Research: Landslide, Liquefaction, and Building Damage via Causal Inference from Satellite Imagery
  Advisor: H. Y. Noh (Stanford), S. Xu (Johns Hopkins), D. Wald (USGS)
- Executive Education Ignite Certificate, Entrepreneurship & Innovation — 2021
- Knight-Hennessy Fellowship, King Global Leadership Program — 2019-22

University of the Philippines Los Baños — Philippines

- BS Civil Engineering, *magna cum laude, college valedictorian, rank 1/341* — 2013-18

Tokyo Metropolitan University — Japan

- Sakura Exchange Program Certificate, Integrated River Engineering Work — 2017

PROFESSIONAL APPOINTMENTS

Oct '22 - Sep '26 — **Doctoral Researcher, UKRI CDT, University of Cambridge, United Kingdom**

- Improved the spatiotemporal resolution of physical exposure models of 44+ Least Developed Countries by 5x finer using geospatial ML on time-series satellite imagery and big datasets of built environment.

- Established a university-industry collaboration with a nonprofit organization and a disaster risk agency in the Philippines for city-scale mapping of multi-hazard urban risk dynamics.

Apr - Jun '26 **Guest Scientist, Digital GreenTalents, German Aerospace Center, Germany**
- Processed 28 GB of census data with over 543 million rows of household data to produce the first nationwide maps of building typology at national, provincial, city, and neighborhood (barangay) scale

Jul - Sep '24 **Guest Scientist, Helmholtz AI, German Aerospace Center, Germany**
- Designed and implemented a deep learning framework that leverages satellite imagery and census constraints to improve spatial disaggregation for building exposure and building typology mapping.

Mar - Aug '22 **Urban Resilience Fellow, Earthquakes and Megacities Initiative, Philippines**
- Developed large-scale seismic risk assessment tools covering 400,000 buildings & 3.2M population for a scenario "Big One" M7.2 earthquake.

Dec '21 - May '22 **Public Policy Graduate Researcher, Stanford University, United States**
- Conducted a probabilistic seismic risk assessment of 23 cities & municipalities in Greater Metro Manila Area and examined their disaster risk management capacities with the country's competitive index.

Mar '21 - Mar '22 **Earthquake Risk Consultant, Stanford University (LBRE), United States**
- Quantified earthquake-induced financial losses and business interruption risks for 750 buildings at Stanford University under seismic scenarios up to magnitude 7.6, and conducted site visits to confirm structural vulnerabilities.

Jun - Sep '21 **Structures & Natural Hazards Researcher, FM Global, United States**
- Improved the firm's library of earthquake design hazard and maps by more than 246%, starting with 130 and ending with 320 maps (or 190 new maps) from over 130 countries.

Jun '20 - Jun '21 **Geospatial & ML Graduate Researcher, Stanford University, United States**
- Implemented a Bayesian causal inference framework leveraging satellite imagery to enhance post-earthquake US Geological Survey landslide, liquefaction, & building damage models by 15x.

Jun - Sep '20 **Structural Retrofit Intern, South Bay Consulting Engineers, United States**
- Produced structural designs and resolved review comments while leading retrofit of a 9,000-sq.ft. two-story home with a weak first story using a cantilevered column system.

Feb - Jun '20 **Housing Research Fellow, New Story Charity, United States**
- Evaluated 7 affordable housing material innovations and led feasibility studies with 20 interdisciplinary fellows across finance, policy, architecture, engineering, and construction.

Jul '18 - Jul '19 **Structural Design Engineer, Arup, Philippines**
- Conducted post-earthquake assessment, performance- and code-based structural peer reviews for multiple 41–50 story high-rise developments.

May - Jul '17 **Planning & Control Engineering Intern, Ayala Group, Philippines**
- Built a VBA-Excel system tracking deliverables of 100+ concurrent projects.

Jun '16 - Jun '17 **Committee Head, Engineering Congress, Philippines**
- Developed event invitation and registration strategies that increased participation fivefold (300 to 1,500 attendees) for major engineering seminars and quiz competitions.

PEER-REVIEWED PUBLICATIONS

[P-1] **Dimasaka, J.**, Bendimerad, F., Tanhueco, R. Geiß, C., and So. E. (2026). Towards Prospective Disaster Risk Management: Mapping Multi-hazard Urban Risk Dynamics Driven by Evolving Exposure and Vulnerability via Earth Observation. *Sustainable Cities and Society* (in review)

[P-2] **Dimasaka, J.**, Geiß, C., and So. E. (2026). GraphVSSM: Graph Variational State-Space Model for Probabilistic Spatiotemporal Inference of Dynamic Exposure and Vulnerability for

Regional Disaster Resilience Assessment. In *Proceedings of the Special Track on AI for Social Impact at the 40th AAAI Conference of Artificial Intelligence*.

[P-3] **Dimasaka, J.,** Geiss, C., Muir-Wood, R., and So, E. (2026). GraphCSVAE: Graph Categorical Structured Variational Autoencoder for Spatiotemporal Auditing of Physical Vulnerability Towards Sustainable Post-Disaster Risk Reduction. *Progress in Disaster Science, Special Issue on AI, Emerging Technologies, and Immersive Solutions for Disaster & Emergency Response.*

[P-4] **Dimasaka, J.**, Geiß, C., and So. E. (2026). DeepC4: Deep Conditional Census-Constrained Clustering for Large-scale Multitask Disaggregation of Urban Morphology. *ISPRS Journal of Photogrammetry and Remote Sensing, Special Issue on Interpretability and Explainability in GeoAI for Geospatial Science and Earth Observations* (in review).

[P-5] **Dimasaka, J.,** Selvakumaran, S., and Marinoni, A. (2024). Enhancing Assessment of Direct and Indirect Exposure of Settlement-Transportation Systems to Mass Movements by Intergraph Representation Learning. *Environmental Research Letters, Focus Collection on Natural Hazards, Disasters, and Extreme Events, 19*(11), 114055.

[P-6] Xu, S., **Dimasaka, J.,** Wald, D., and Noh, H.Y. (2022). Seismic Multi-hazard and Impact
[P-7] Estimation via Causal Inference from Satellite Imagery. *Nature Communications, 13*(1), 7793.

JOURNAL MANUSCRIPTS IN PREPARATION

[M-1] Quantifying the Regional Dynamics and Redistribution of Physical Vulnerability in Least Developed Countries. *(Target Journal: Global Environmental Change).*

[M-2] From Census to Maps: A Nationwide Three-Decade Analysis of Exposure, Vulnerability, and Risk of the Philippines (1990–2020). *(Target Journal: Nature Communications)*

CONFERENCE PRESENTATIONS

[C-1] Spatial Disaggregation and Temporal Projection of Building Exposure and Physical Vulnerability using Deep Constrained Clustering & Probabilistic Graph Deep Learning. *GEM Conference: From Faults to Future Scenarios, Dynamic Exposure Modelling*, Croatia, Jun 24, 2026.

[C-2] METEOR 2.5D: An Open Geospatial Dataset of the Spatiotemporal Evolution of Physical Vulnerability in UN-recognized Least Developed Countries (as of 2020) at Five-year Intervals, 1975-2030. *EGU General Assembly 2026*, Austria, May 5, 2026.

[C-3] A Data-Driven Probabilistic Approach to Regional Dynamics of Building Exposure and Physical Vulnerability Towards Global Disaster Risk Quantification Audit. *AGU Annual Meeting 2025,* United States, Dec 16, 2025.

[C-4] Modernizing Quezon City's Building Risk Data with AI and Earth Observation. *20th Assoc. of Pacific Rim Universities - Multi-Hazards Conference & Symposium*, Philippines, Nov 28, 2025.

[C-5] Spatial Disaggregation of Rwandan Building Exposure and Vulnerability via Weakly Supervised Conditional Census-Constrained Clustering (C4) using Earth Observation Data. *AGU Annual Meeting 2024*, United States, Dec 10, 2024.

[C-6] Global Mapping of Exposure & Physical Vulnerability Dynamics in Least Developed Countries using Remote Sensing & Machine Learning. *12th Int'l Conference on Learning Representation (ICLR), 2nd Machine Learning for Remote Sensing Workshop*, Austria, May 11, 2024.

[C-7] Near-real-time Country-wide Estimation of Susceptibility and Settlement Exposure from Norwegian Mass Movements via Inter-graph Representation Learning. *AGU Annual Meeting 2023*, United States, Dec 14, 2023.

[C-8] Improving Post-earthquake Disaster Response using Bayesian-Updated Ground Failure Models with Satellite Imagery. *20th Assoc. of Structural Engineers of the Philippines (ASEP) International Convention*, Philippines, May 21, 2021.

INVITED TALKS

[I-1] Near-real-time Country-wide Estimation of Susceptibility and Settlement Exposure from Norwegian Mass Movements via Intergraph Representation Learning. *Innovations Group WG5 of LandAware, an int'l network on early warning systems,* Online, Jun 25, 2024.

[I-2] Computing in Civil Eng'g. *University of the Philippines Los Baños*, PH, Sep 29, 2023.

[I-3] Mapping Global Disaster Risk using AI and Earth Observation Data: Examples from Climate-Induced Mass Movements and Seismic Multi-Hazard Impact Assessment. *Moody's Risk Management Solutions – Model Development Team*, London, UK, Jun 21, 2023.

TECHNICAL REPORTS

[T-1] Tanhueco, R.M., Bendimerad, F., Khazai, B., Tibig, L., **Dimasaka, J.**, and Luzon, P.K. (2022). Climate and Disaster Risk Assessment Report for Quezon City: Climate Change, Earthquake, Flood, and Landslide Hazards, including Identification of Hotspot Barangays. Quezon City Government and Earthquakes & Megacities Initiative.

[T-2] Bendimerad, F., Canney, N., and **Dimasaka, J.** (2021). Seismic Risk and Loss Estimation of Stanford University Buildings.

OPEN-SOURCE DATA INITIATIVES

[O-1] METEOR 2.5D: An Open Geospatial Dataset of Spatiotemporal Evolution of Physical Vulnerability in UN-recognized Least Developed Countries (as of 2020) at Five-year Intervals, 1975-2030. ***doi.org/10.5281/zenodo.16608380* | *doi.org/10.5281/zenodo.16695059***

[O-2] A Local-Scale Dataset of Annual Spatiotemporal Maps of Physical Vulnerability in the Cyclone-Impacted Coastal Khurushkul Community (Bangladesh) and Mudslide-Affected Freetown (Sierra Leone) (2016–2023). ***doi.org/10.5281/zenodo.16656471***

[O-3] A City-Scale Dataset of Annual Spatiotemporal Maps of Building Exposure and Physical Vulnerability in Quezon City, Philippines (2016–2030). ***doi.org/10.5281/zenodo.16655873***

[O-4] OpenSendaiBench: A Benchmark Dataset of Building Exposure and Vulnerability Dynamics for EO-based Auditing of Global Disaster Risk. ***doi.org/10.5281/zenodo.10840484***

TEACHING

2025 Presenter, Advanced Animations for Scientific Oral Presentation, *University of Cambridge*

2023 Demonstrator, Geospatial Data Visualization (Basics of GIS), *University of Cambridge*

VOLUNTEERING

2026 Reviewer, *International Journal of Digital Earth*
Reviewer, *4th ML for Remote Sensing Workshop, ICLR*

2025 Reviewer, *AAAI Special Track on AI for Social Impact & AI for Urban Planning Workshop*
Reviewer, *Updated UNDRR-International Science Council Hazard Information Profiles*
Lead, *Mw-7.7 Seismic Impact Assessment in Myanmar & Thailand using Remote Sensing Data*

2024 Reviewer, *American Geophysical Union Annual Meeting 2024, OSPA*
Reviewer, *GeoAI for a Sustainable Future in Africa Workshop*
Reviewer, *Cambridge Journal of Science and Policy*
Exhibitor, *Cambridge Festival*

2023-25 Admission Ambassador for Scholar Q&A, *Stanford Knight-Hennessy Scholars Program*

2023 Reviewer, *American Geophysical Union Annual Meeting 2023, OSPA*
Participant, *Joint Workshop of Disaster Research Group & UK Alliance for Disaster Research*

2022 Reviewer, *Intersect: The Stanford Journal of Science, Technology, and Society*

2020 Fellow, *New Story Charity (Affordable Housing Solutions)*

2017 Lead Organizer, *12th Engineering Congress (Philippine, National)*

2014-22 Lead Coordinator and Interviewer, *Ramar Scholarship Foundation for Low-Income Students*

RECOGNITIONS

2026	British Council Early Career Fellowship on AI Applications	£	57,000
2026	EGU Outstanding Student & PhD candidate Presentation, *European Geoscience Union*		
2025	Digital GreenTalents, *Federal Ministry of Research, Technology & Space, DE*	€	5,850
2025	Best Overall Paper Conference, *8th Int'l Disaster & Risk Conference, CY*		

2024	Helmholtz Visiting Researcher Grant, *Helmholtz Association, DE*	€	13,040
2023	AGU Outstanding Student Presentation Award (OSPA), *American Geophysical Union*		
2022-26	UKRI CDT Studentship, *Eng'g & Physical Sciences Research Council, UK*	£	106,000
	Departmental Award, *Earth Sciences, University of Cambridge, UK*	£	106,000
2019-22	Knight-Hennessy Graduate Fellowship, *Stanford University, US*	$	240,000
2018	Int'l Publication Award, *University of the Philippines, PH*		
2018	BPI-DOST Science Award, *BPI Foundation and Dept of Science & Tech, PH*		
2017	Best Scientific Oral Presentation, *ASEAN-CAFMN, PH*		
2017	3rd Place, Most Outstanding Civil Eng'g Student National Award, *PICE, PH*		
2013-17	DOST RA 7687 & RSFI Undergraduate Scholarships, *PH*	₱	547,000

REFERENCES

Emily So MEng PhD CEng FICE
Director, Cambridge University Centre
for Risk in the Built Environment (CURBE),
United Kingdom

Fouad Bendimerad PhD
Chairman,
Earthquakes & Megacities Initiative,
Philippines

Susu Xu PhD
Assistant Professor, Department of
Civil and Systems Engineering,
Johns Hopkins University,
United States

Marish Madlangbayan PhD
Former Head and Professor,
Department of Civil Engineering,
University of the Philippines Los Baños,
Philippines

Christian Geiß PhD
Head, Georisks, German Aerospace Center
and University of Bonn,
Germany

David Wald PhD
Seismologist,
U.S. Geological Survey (USGS),
United States

Hae Young Noh PhD
Associate Professor, Department of
Civil and Environmental Engineering,
Stanford University,
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Professor, Institute of Physics,
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